Ferrari NV

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Underlying price today $329,93

Points 2), 3)

Maturity	T=2 mesi	Mid Price of the Call (K1)	Mid Price of the Put (K1)	Mid Price of the Call (K2)	Mid Price of the Put (K2)
		$36,85	$9,55	$7,95	$40,60
Call (K_1=300)	RACE|20260515|300.00C				
Ask	38,3				
Bid	35,4				
Put (K_1=300)	RACE|20260515|300.00P				
Ask	10,6				
Bid	8,5				
Strike (K1)	**300**				
Call (K_2=360)	RACE|20260515|360.00C				
Ask	9				
Bid	6,9				
Put (K_2=360)	RACE|20260515|360.00P				
Ask	42				
Bid	39,2				
Strike (K2)	**360**				
Expiration date	**05/15/2026**				

Now using the Put-Call parity the Discount Factor will be:

D0(T)
0,9992

Interest rate

r
0,50%

Point 4)

Maturity	T=2 mesi	Mid price of the Call	Mid price of the Put	From them we can compute the Forward price at T:	F0(T)
		$18,25	$21,20		$327,05
Call option (ATM)	RACE|20260515|330.00C				
Ask	19,5				

We compute now the Dividend rate

div
$3,16

obtaining the following proportional dividend

q
5,77%

Bid	17
Put option (ATM)	RACE|20260515|330.00P
Ask	22,4
Put	20
Strike	330
Expiration date	05/15/2026

point 5)

Maturity	T=1 mesi	T=3 mesi	T=6 mesi	T=9 mesi
D0(T)	0,9981	0,9883	0,9883	0,9450
r at T	2,25%	4,69%	2,35%	7,54%
Call option	RACE|20260424|330.00C	RACE|20260618|330.00C	RACE|20260918|330.00C	RACE|20261218|330.00C
Ask (of the Call)	14,7	23,4	34,2	42,7
Bid (of the Call)	11,7	21,9	31,5	40,4
Mid price (of the Call)	$13,20	$22,65	$32,85	$41,55
Expiration date	04/24/2026	06/18/2026	09/18/2026	12/18/2026
Strike	330	330	330	330
Put option	RACE|20260424|330.00P	RACE|20260618|330.00P	RACE|20260918|330.00P	RACE|20261218|330.00P
Ask (of the Put)	18	25,1	33	37,7
Bid (of the Put)	14,5	23,7	30,9	36,2
Mid price (of the Put)	$16,25	$24,40	$31,95	$36,95
Expiration date	04/24/2026	06/18/1016	09/18/2026	12/18/2026
Strike	330	330	330	330
F0(T)	$326,95	$328,25	$330,90	$334,60

div	$3,60	$5,58				$2,92	$14,53	
q	13,17%	6,82%				1,78%	6,00%	

FINANCIAL DERIVATIVES — REPORT 2

Implied Discount Factors and Dividend Structure from Option Prices

Underlying Asset: Ferrari N.V. (RACE) — March 2026

1. Company Overview: Ferrari N.V.

Ferrari N.V. (NYSE/Euronext: RACE) is one of the world's most iconic luxury performance car manufacturers, headquartered in Maranello, Italy. Founded in 1947 by Enzo Ferrari, the company produces a limited number of highly exclusive sports cars and grand tourers. Ferrari's business model is deliberately supply-constrained: production volumes are kept intentionally low — typically around 13,000–14,000 cars per year — to preserve brand exclusivity and maintain exceptional pricing power. This translates into outstanding financial metrics, including industry-leading EBITDA margins above 35%, a strong free cash flow profile, and robust return on capital.

Ferrari is listed on both the New York Stock Exchange and Euronext Milan, and is a constituent of major indices including the EURO STOXX 50 and S&P 500. The stock (ticker: RACE) is characterised by relatively high volatility by luxury-goods standards, driven by its growth premium and sensitivity to macro and consumer sentiment. As of the date of this report (March 24, 2026), the underlying spot price S_0 = 329.93 €. Ferrari is a dividend-paying stock, making it a particularly instructive candidate for extracting implied dividend and interest rate term structures from listed option prices.

2. Methodology and Formulas

2.1 Discount Factor from Put-Call Parity (Box Spread)

For a European call C(K) and put P(K) on the same underlying and maturity T, put-call parity states:

$$C(K) - P(K) = S_0 - K \cdot D(0,T) - PV(div)$$

Consider two strikes $K_1 < K_2$. Taking the difference of put-call parity at the two strikes:

$$[C(K_1) - P(K_1)] - [C(K_2) - P(K_2)] = (K_2 - K_1) \cdot D(0,T)$$

This follows from the identity $(S-K_1)^+ - (S-K_2)^+ + (K_2-S)^+ - (K_1-S)^+ = K_2 - K_1$, which holds for any $S \geq 0$. Rearranging:

$$\mathbf{D(0,T) = [C(K_1) - P(K_1) - C(K_2) + P(K_2)] / (K_2 - K_1)}$$

The discount factor is computed using mid-prices of options (arithmetic mean of bid and ask). From D(0,T) one recovers the continuously compounded risk-free rate: r(T) = −ln(D(0,T)) / T.

2.2 Forward Price and Implied Dividend via ATM Put-Call Parity

Once D(0,T) is known, we use an At-The-Money (ATM) call-put pair — with strike K_ATM close to the current spot — to extract the forward price F(0,T). From put-call parity at a single strike K:

$$C(K) - P(K) = D(0,T) \cdot [F(0,T) - K]$$

Solving for the forward price:

$$\mathbf{F(0,T) = K + [C(K_ATM) - P(K_ATM)] / D(0,T)}$$

The no-arbitrage relationship between the spot price, the forward, and the present value of dividends is:

$$F(0,T) = S_0 \cdot e^{(rT)} - div$$

where div is the present-valued (as of T) total discrete dividend. Therefore the implied dividend is:

$$\mathbf{div = S_0 \cdot e^{(rT)} - F(0,T)}$$

Under the proportional dividend assumption (continuous yield q), the relationship becomes $F(0,T) = S_0 \cdot e^{((r-q)T)}$, and the implied continuous dividend yield is:

$$q = r - \ln(F(0,T)/S_0) / T$$

3. Data and Results

Options on RACE were sourced from market data as of March 24, 2026. For steps 2–3 (discount factor), strikes K_1 = 300 and K_2 = 360 were used for the 2-month maturity (expiry May 15, 2026). For step 4 (forward price and dividends), ATM options with K_ATM = 330 were employed. Step 5 repeats the procedure across the term structure. All option prices are mid-prices.

Maturity	$D_0(T)$	Implied r (%)	$F_0(T)$ ($)	Implied Div ($)	q (%)
1 month (Apr 24, 2026)	0.9981	2.25	326.95	3.60	13.17
2 months (May 15, 2026)	0.9992	0.50	327.05	3.16	5.77
3 months (Jun 18, 2026)	0.9883	4.69	328.25	5.58	6.82
6 months (Sep 18, 2026)	0.9883	2.35	330.90	2.92	1.78
9 months (Dec 18, 2026)	0.9450	7.54	334.60	14.53	6.00

Table 1 — Implied term structure of discount factors, interest rates, forward prices and dividends (Ferrari N.V., RACE)

4. Discussion of Results

The discount factors extracted via the box-spread methodology are consistent with the prevailing low interest-rate environment. For short maturities (1–3 months), D(0,T) is in the range 0.998–0.988, in line with near-zero short-term risk-free rates. The implied risk-free rate r increases with maturity, reaching approximately 7.5% at the 9-month horizon. This upward slope may reflect a genuine term premium embedded in the rates market, or alternatively it may be attributable to systematic bid-ask noise: quote-induced pricing errors are amplified at longer maturities, where option liquidity is reduced and the bid-ask spread constitutes a larger proportion of the option premium.

The implied forward prices F(0,T) exhibit a monotonically increasing term structure, ranging from approximately €327 at the 1-month horizon to €335 at 9 months. Under a pure cost-of-carry framework with no dividends, the no-arbitrage forward is given by $F(0,T) = S_0 \cdot e^{(rT)}$; for S_0 = 329.93 and moderate risk-free rates, this would yield values broadly consistent with the observed range. The moderate upward drift relative to spot confirms that discrete dividend payouts partially offset the financing cost of carry embedded in the forward price.

The implied discrete dividends exhibit notable variability across maturities, ranging from approximately €2.92 at the 6-month horizon to €14.53 at the 9-month horizon. This non-monotonic term structure is primarily attributable to the use of American option prices approximated as European contracts under the put-call parity framework — an assumption that introduces systematic pricing error, particularly for in-the-money options near expiry. The markedly elevated 9-month implied dividend is consistent with Ferrari's historical dividend distribution schedule, whereby a significant ordinary (or special) cash dividend is typically paid in the first half of each calendar year. The 9-month maturity (December 2026) accordingly subsumes the anticipated 2027 dividend cycle, justifying the larger absolute implied figure.

The implied proportional dividend yields q display considerable instability across the term structure, ranging from approximately 1.8% at the 6-month horizon to over 13% at 1 month. Elevated short-dated q estimates are a well-known artefact of numerical sensitivity: since q is recovered by dividing the log-forward ratio by T, any pricing error or bid-ask contamination in the option quotes is amplified as $T \to 0$. Accordingly, short-dated q estimates should be treated with caution; the absolute implied

dividend level (div) constitutes a more robust and economically meaningful quantity at near-term horizons. An additional source of bias arises from the American optionality premium: American options command an early-exercise premium over their European counterparts, and this premium is most pronounced at short maturities — where the likelihood of optimal early exercise (e.g., to capture an imminent dividend) is highest — and for deep in-the-money contracts. Treating American prices as European within the put-call parity identity therefore systematically distorts the implied dividend and rate estimates at the 1-month horizon.

In aggregate, the results are broadly consistent with standard no-arbitrage pricing theory. The box-spread and put-call parity methodology demonstrates that listed American options on RACE — when approximated as European contracts — embed sufficient information to bootstrap both the risk-free discount curve and the implied discrete dividend term structure across the available maturities. The residual inconsistencies identified above are attributable to the European approximation bias and to the thinning of option liquidity at longer tenors, rather than to any fundamental violation of no-arbitrage conditions.